

```
ss.sql lab4.sql x
lab4.sql
1 -- 1
2 SELECT UPPER(airline_name)
3 FROM airline;
4
5 -- -- 2
6 -- SELECT REPLACE(airline_name, 'Air', 'Aero')
7 -- FROM airline;
8
9 -- -- 3
10 -- SELECT flight_id
11 -- FROM flights
12 -- WHERE airline_id IN (1, 2);
13
14 -- -- 4
15 -- SELECT *
16 -- FROM airport
17 -- WHERE airport_name LIKE '%Reginal%'
18 -- AND airport_name LIKE '%Air%';
19

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
macbookair@Aknur:~$ practice databasel.sql % psql lab3
psql (14.22 (Homebrew), server 18.3 (Homebrew))
WARNING: psql major version 14, server major version 18.
Some psql features might not work.
Type "help" for help.

lab3=# \i lab4.sql
upper
-----
AIR FRANCE
KAZ AIR
TAP AIR
POLISH AIR
TURKISH AIRLINES
LUFTHANSA
(6 rows)

lab3=#
```

```
ss.sql lab4.sql x
lab4.sql
60 -- -- 12
61 --SELECT *
62 --FROM airline
63 --WHERE airline_code ~ '[0-9]';
64
65 -- -- 13
66 SELECT *
67 FROM airline
68 ORDER BY created_at DESC
69 LIMIT 5;
70
71 -- -- 14
72 --SELECT *
73 --FROM baggage_check
74 --WHERE booking_id BETWEEN 200 AND 300
75 --AND check_result <> 'Checked';

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
macbookair@Aknur:~$ practice databasel.sql % psql lab3
airline_id | airline_code | airline_name | airline_country | created_at | updated_at
-----
6 | LH | Lufthansa | Germany | 2024-06-01 00:00:00 | 2024-06-01 00:00:00
5 | TK | Turkish Airlines | Turkey | 2024-05-01 00:00:00 | 2024-05-01 00:00:00
4 | PL4 | Polish Air | Poland | 2024-03-10 00:00:00 | 2024-03-10 00:00:00
2 | K2 | Kaz Air | Kazakhstan | 2024-02-15 00:00:00 | 2024-02-15 00:00:00
1 | A1 | Air France | France | 2024-01-10 00:00:00 | 2024-01-10 00:00:00
(5 rows)

:
airline_id | airline_code | airline_name | airline_country | created_at | updated_at
-----
6 | LH | Lufthansa | Germany | 2024-06-01 00:00:00 | 2024-06-01 00:00:00
5 | TK | Turkish Airlines | Turkey | 2024-05-01 00:00:00 | 2024-05-01 00:00:00
4 | PL4 | Polish Air | Poland | 2024-03-10 00:00:00 | 2024-03-10 00:00:00
2 | K2 | Kaz Air | Kazakhstan | 2024-02-15 00:00:00 | 2024-02-15 00:00:00
1 | A1 | Air France | France | 2024-01-10 00:00:00 | 2024-01-10 00:00:00
(5 rows)

:
airline_id | airline_code | airline_name | airline_country | created_at | updated_at
-----
6 | LH | Lufthansa | Germany | 2024-06-01 00:00:00 | 2024-06-01 00:00:00
5 | TK | Turkish Airlines | Turkey | 2024-05-01 00:00:00 | 2024-05-01 00:00:00
4 | PL4 | Polish Air | Poland | 2024-03-10 00:00:00 | 2024-03-10 00:00:00
```

```
ss.sql lab4.sql x
lab4.sql
1 -- -- 1
2 -- SELECT UPPER(airline_name)
3 -- FROM airline;
4
5 -- -- 2
6 SELECT REPLACE(airline_name, 'Air', 'Aero')
7 FROM airline;
8
9 -- -- 3
10 -- SELECT flight_id
11 -- FROM flights
12 -- WHERE airline_id IN (1, 2);
13
14 -- -- 4
15 -- SELECT *
16 -- FROM airport
17 -- WHERE airport_name LIKE '%Reginal%'
18 -- AND airport_name LIKE '%Air%';
19

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
macbookair@Aknur-MacBook practice database1.sql % psql lab3
psql (14.22 (Homebrew), server 18.3 (Homebrew))
WARNING: psql major version 14, server major version 18.
Some psql features might not work.
Type "help" for help.

lab3=# \i lab4.sql
replace
-----
Aero France
Kaz Aero
TAP Aero
Polish Aero
Turkish Aerolines
Lufthansa
(6 rows)

lab3=#
```

```
ss.sql lab4.sql x
lab4.sql
1 -- -- 1
2 -- SELECT UPPER(airline_name)
3 -- FROM airline;
4
5 -- -- 2
6 -- SELECT REPLACE(airline_name, 'Air', 'Aero')
7 -- FROM airline;
8
9 -- -- 3
10 SELECT flight_id
11 FROM flights
12 WHERE airline_id IN (1, 2);
13
14 -- -- 4
15 -- SELECT *
16 -- FROM airport
17 -- WHERE airport_name LIKE '%Reginal%'
18 -- AND airport_name LIKE '%Air%';
19

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
macbookair@Aknur-MacBook practice database1.sql % psql lab3
psql (14.22 (Homebrew), server 18.3 (Homebrew))
WARNING: psql major version 14, server major version 18.
Some psql features might not work.
Type "help" for help.

lab3=# \i lab4.sql
flight_id
-----
1
2
(2 rows)

lab3=#
```

```
ss.sql lab4.sql x
lab4.sql
5 -- -- 2
6 -- SELECT REPLACE(airline_name, 'Air', 'Aero')
7 -- FROM airline;
8
9 -- -- 3
10 -- SELECT flight_id
11 -- FROM flights
12 -- WHERE airline_id IN (1, 2);
13
14 -- 4
15 SELECT *
16 FROM airport
17 WHERE airport_name LIKE '%Reginal%'
18 AND airport_name LIKE '%Air%';
19
20 -- -- 5
21 -- SELECT first_name, last_name,
22 -- TO_CHAR(date_of_birth, 'Month DD, YYYY')
23 -- FROM passengers;

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
macbookair@Aknur:~$ practice database1.sql % psql lab3
psql (14.22 (Homebrew), server 18.3 (Homebrew))
WARNING: psql major version 14, server major version 18.
Some psql features might not work.
Type "help" for help.

lab3=# \i lab4.sql
airport_id | airport_name | country | state | city | created_at | updated_at
-----
1 | Astana Reginal Air | Kazakhstan | Astana | Astana | 2023-01-01 00:00:00 | 2023-01-01 00:00:00
3 | London Reginal Air | UK | London | London | 2023-01-01 00:00:00 | 2023-01-01 00:00:00
(2 rows)

(END)
airport_id | airport_name | country | state | city | created_at | updated_at
-----
1 | Astana Reginal Air | Kazakhstan | Astana | Astana | 2023-01-01 00:00:00 | 2023-01-01 00:00:00
3 | London Reginal Air | UK | London | London | 2023-01-01 00:00:00 | 2023-01-01 00:00:00
(2 rows)
```

```
ss.sql lab4.sql x
lab4.sql
15 -- SELECT *
16 -- FROM airport
17 -- WHERE airport_name LIKE '%Reginal%'
18 -- AND airport_name LIKE '%Air%';
19
20 -- 5
21 SELECT first_name, last_name,
22 TO_CHAR(date_of_birth, 'Month DD, YYYY')
23 FROM passengers;
24
25 -- -- 6
26 -- SELECT flight_id
27 -- FROM flights
28 -- WHERE act_arrival_time > sch_arrival_time;
29
30 -- -- 7
31 -- SELECT flight_id, sch_arrival_time, act_arrival_time
32 -- FROM flights
33 -- WHERE act_arrival_time > sch_arrival_time;

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
macbookair@Aknur:~$ practice database1.sql % psql lab3
psql (14.22 (Homebrew), server 18.3 (Homebrew))
WARNING: psql major version 14, server major version 18.
Some psql features might not work.
Type "help" for help.

lab3=# \i lab4.sql
first_name | last_name | to_char
-----
Ali | Amanov | April 10, 2001
Aruzhan | Samatova | August 15, 2004
Dias | Bekov | November 20, 1999
Dana | Aliyeva | May 12, 2006
Arman | Serikov | February 01, 2002
(5 rows)

lab3=#
```

```
ss.sql lab4.sql x
lab4.sql
22 -- TO_CHAR(date_of_birth, 'Month DD, YYYY')
23 -- FROM passengers;
24
25 -- 6
26 SELECT flight_id
27 FROM flights
28 WHERE act_arrival_time > sch_arrival_time;
29
30 -- -- 7
31 -- SELECT flight_id, sch_arrival_time, act_arrival_time
32 -- FROM flights
33 -- WHERE act_arrival_time > sch_arrival_time;
34
35 -- -- 8
36 -- SELECT *
37 -- FROM airline
38 -- WHERE airline_country IN ('France', 'Portugal', 'Poland')
39 -- AND created_at >= '2023-11-01'
40 -- AND created_at < '2024-04-01';
41

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
macbookair@Aknur-MacBook practice databasel.sql % psql lab3
psql (14.22 (Homebrew), server 18.3 (Homebrew))
WARNING: psql major version 14, server major version 18.
Some psql features might not work.
Type "help" for help.

lab3=# \i lab4.sql
flight_id
-----
1
3
(2 rows)

lab3=#
```

```
ss.sql lab4.sql x
lab4.sql
20 -- -- 5
21 -- SELECT first_name, last_name,
22 -- TO_CHAR(date_of_birth, 'Month DD, YYYY')
23 -- FROM passengers;
24
25 -- 6
26 --SELECT flight_id
27 --FROM flights
28 --WHERE act_arrival_time > sch_arrival_time;
29
30 -- 7
31 SELECT flight_id, sch_arrival_time, act_arrival_time
32 FROM flights
33 WHERE act_arrival_time > sch_arrival_time;
34
35 -- -- 8
36 -- SELECT *
37 -- FROM airline
38 -- WHERE airline_country IN ('France', 'Portugal', 'Poland')

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
macbookair@Aknur-MacBook practice databasel.sql % psql lab3
psql (14.22 (Homebrew), server 18.3 (Homebrew))
WARNING: psql major version 14, server major version 18.
Some psql features might not work.
Type "help" for help.

lab3=# \i lab4.sql
flight_id | sch_arrival_time | act_arrival_time
-----
1 | 2024-05-01 12:00:00 | 2024-05-01 12:30:00
3 | 2024-05-03 15:00:00 | 2024-05-03 15:45:00
(2 rows)

lab3=#
```

```
ss.sql lab4.sql x
lab4.sql
1 -- -- 1
2 -- SELECT UPPER(airline_name)
3 -- FROM airline;
4
5 -- -- 2
6 -- SELECT REPLACE(airline_name, 'Air', 'Aero')
7 -- FROM airline;
8
9 -- -- 3
10 -- SELECT flight_id
11 -- FROM flights
12 -- WHERE airline_id IN (1, 2);
13
14 -- -- 4
15 -- SELECT *
16 -- FROM airport
17 -- WHERE airport_name LIKE '%Reginal%'
18 -- AND airport_name LIKE '%Air%';
19

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
macbookair@Aknur-MacBook practice databasel.sql % psql lab3
psql (14.22 (Homebrew), server 18.3 (Homebrew))
WARNING: psql major version 14, server major version 18.
Some psql features might not work.
Type "help" for help.

lab3=# \i lab4.sql
airline_id | airline_code | airline_name | airline_country | created_at | updated_at
-----
1 | A1 | Air France | France | 2024-01-10 00:00:00 | 2024-01-10 00:00:00
3 | TP | TAP Air | Portugal | 2023-12-20 00:00:00 | 2023-12-20 00:00:00
4 | PL4 | Polish Air | Poland | 2024-03-10 00:00:00 | 2024-03-10 00:00:00
(3 rows)

(END)
airline_id | airline_code | airline_name | airline_country | created_at | updated_at
-----
1 | A1 | Air France | France | 2024-01-10 00:00:00 | 2024-01-10 00:00:00
3 | TP | TAP Air | Portugal | 2023-12-20 00:00:00 | 2023-12-20 00:00:00
4 | PL4 | Polish Air | Poland | 2024-03-10 00:00:00 | 2024-03-10 00:00:00
(3 rows)
```

```
ss.sql lab4.sql x
lab4.sql
34
35 -- -- 8
36 --SELECT *
37 --FROM airline
38 --WHERE airline_country IN ('France', 'Portugal', 'Poland')
39 --AND created_at >= '2023-11-01'
40 --AND created_at < '2024-04-01';
41
42 -- -- 9
43 SELECT *
44 FROM baggage
45 WHERE weight_in_kg > 25
46 ORDER BY weight_in_kg DESC
47 LIMIT 3;
48
49 -- -- 10
50 -- SELECT first_name, last_name
51 -- FROM passengers
52 -- ORDER BY date_of_birth DESC

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
macbookair@Aknur-MacBook practice databasel.sql % psql lab3
psql (14.22 (Homebrew), server 18.3 (Homebrew))
WARNING: psql major version 14, server major version 18.
Some psql features might not work.
Type "help" for help.

lab3=# \i lab4.sql
baggage_id | weight_in_kg | created_at | updated_at | booking_id
-----
3 | 41.20 | 2024-04-03 00:00:00 | 2024-04-03 00:00:00 | 250
5 | 35.70 | 2024-04-05 00:00:00 | 2024-04-05 00:00:00 | 301
1 | 32.50 | 2024-04-01 00:00:00 | 2024-04-01 00:00:00 | 205
(3 rows)

(END)
baggage_id | weight_in_kg | created_at | updated_at | booking_id
-----
3 | 41.20 | 2024-04-03 00:00:00 | 2024-04-03 00:00:00 | 250
5 | 35.70 | 2024-04-05 00:00:00 | 2024-04-05 00:00:00 | 301
1 | 32.50 | 2024-04-01 00:00:00 | 2024-04-01 00:00:00 | 205
(3 rows)
```

```
ss.sql lab4.sql x
lab4.sql
43 --SELECT *
44 --FROM baggage
45 --WHERE weight_in_kg > 25
46 --ORDER BY weight_in_kg DESC
47 --LIMIT 3;
48
49 -- -- 10
50 SELECT first_name, last_name
51 FROM passengers
52 ORDER BY date_of_birth DESC
53 LIMIT 1;
54
55 -- -- 11
56 --SELECT booking_platform, MIN(ticket_price)
57 --FROM booking
58 --GROUP BY booking_platform;

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
macbookair@Aknur-MacBook practice databasel.sql % \pset pager off
zsh: command not found: pset
macbookair@Aknur-MacBook practice databasel.sql % \i lab4.sql
zsh: command not found: i
macbookair@Aknur-MacBook practice databasel.sql % \pset pager off
zsh: command not found: pset
macbookair@Aknur-MacBook practice databasel.sql % \i lab4.sql
zsh: command not found: i
macbookair@Aknur-MacBook practice databasel.sql % psql lab3
psql (14.22 (Homebrew), server 18.3 (Homebrew))
WARNING: psql major version 14, server major version 18.
Some psql features might not work.
Type "help" for help.

lab3=# \i lab4.sql
first_name | last_name
-----
Dana       | Aliyeva
(1 row)

lab3=#
```

```
ss.sql lab4.sql x
lab4.sql
45 --WHERE weight_in_kg > 25
46 --ORDER BY weight_in_kg DESC
47 --LIMIT 3;
48
49 -- -- 10
50 --SELECT first_name, last_name
51 --FROM passengers
52 --ORDER BY date_of_birth DESC
53 --LIMIT 1;
54
55 -- -- 11
56 SELECT booking_platform, MIN(ticket_price)
57 FROM booking
58 GROUP BY booking_platform;
59
60 -- -- 12
61 -- SELECT *
62 -- FROM airline
63 -- WHERE airline_code ~ '[0-9]';

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
macbookair@Aknur-MacBook practice databasel.sql % psql lab3
psql (14.22 (Homebrew), server 18.3 (Homebrew))
WARNING: psql major version 14, server major version 18.
Some psql features might not work.
Type "help" for help.

lab3=# \i lab4.sql
booking_platform | min
-----
Website          | 15000.00
Mobile App       | 17000.00
Kaspi            | 18000.00
(3 rows)

lab3=#
```

```
ss.sql lab4.sql x
lab4.sql
50 --SELECT first_name, last_name
51 --FROM passengers
52 --ORDER BY date_of_birth DESC
53 --LIMIT 1;
54
55 -- -- 11
56 --SELECT booking_platform, MIN(ticket_price)
57 --FROM booking
58 --GROUP BY booking_platform;
59
60 -- -- 12
61 SELECT *
62 FROM airline
63 WHERE airline_code ~ '[0-9]';
64
65 -- -- 13
66 -- SELECT *
67 -- FROM airline
68 -- ORDER BY created_at DESC

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
macbookair@Aknur—MacBook practice databasel.sql % psql lab3
psql (14.22 (Homebrew), server 18.3 (Homebrew))
WARNING: psql major version 14, server major version 18.
Some psql features might not work.
Type "help" for help.

lab3=# \i lab4.sql
 airline_id | airline_code | airline_name | airline_country | created_at | updated_at
-----
1 | A1 | Air France | France | 2024-01-10 00:00:00 | 2024-01-10 00:00:00
2 | K2 | Kaz Air | Kazakhstan | 2024-02-15 00:00:00 | 2024-02-15 00:00:00
4 | PL4 | Polish Air | Poland | 2024-03-10 00:00:00 | 2024-03-10 00:00:00
(3 rows)

(END)
 airline_id | airline_code | airline_name | airline_country | created_at | updated_at
-----
1 | A1 | Air France | France | 2024-01-10 00:00:00 | 2024-01-10 00:00:00
2 | K2 | Kaz Air | Kazakhstan | 2024-02-15 00:00:00 | 2024-02-15 00:00:00
4 | PL4 | Polish Air | Poland | 2024-03-10 00:00:00 | 2024-03-10 00:00:00
(3 rows)
```

```
ss.sql lab4.sql x
lab4.sql
67 --FROM airline
68 --ORDER BY created_at DESC
69 --LIMIT 5;
70
71 -- -- 14
72 --SELECT *
73 --FROM baggage_check
74 --WHERE booking_id BETWEEN 200 AND 300
75 --AND check_result <> 'Checked';
76
77 -- -- 15
78 SELECT *
79 FROM baggage_check
80 WHERE DATE_TRUNC('month', updated_at)
81 | = DATE_TRUNC('month', created_at)
82 AND updated_at < created_at;

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
macbookair@Aknur—MacBook practice databasel.sql % psql lab3
lab3=# \i lab4.sql
 baggage_check_id | check_result | created_at | updated_at | booking_id | passenger_id
-----
1 | Not checked | 2024-04-10 12:00:00 | 2024-04-10 11:00:00 | 205 | 1
(1 row)

(END)
 baggage_check_id | check_result | created_at | updated_at | booking_id | passenger_id
-----
1 | Not checked | 2024-04-10 12:00:00 | 2024-04-10 11:00:00 | 205 | 1
(1 row)

~
~
~
```

ss.sqllab4.sql

lab4.sql

61 --SELECT *
62 --FROM airline
63 --WHERE airline_code ~ '[0-9]';
64
65 -- -- 13
66 --SELECT *
67 --FROM airline
68 --ORDER BY created_at DESC
69 --LIMIT 5;
70
71 -- -- 14
72 SELECT *
73 FROM baggage_check
74 WHERE booking_id BETWEEN 200 AND 300
75 AND check_result <> 'Checked';
76
77 -- -- 15
78 -- SELECT *
79 -- FROM baggage_check
80 -- WHERE DATE_TRUNC('month', updated_at)
81 -- = DATE_TRUNC('month', created_at)
82 -- AND updated_at < created_at;

PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS

psql + - ...

macbookair@Aknur--MacBook practice databasel.sql % psql lab3

lab3=# \i lab4.sql
baggage_check_id | check_result | created_at | updated_at | booking_id | passenger_id

1 | Not checked | 2024-04-10 12:00:00 | 2024-04-10 11:00:00 | 205 | 1
3 | Not checked | 2024-04-12 13:00:00 | 2024-04-12 14:00:00 | 250 | 3
4 | Rejected | 2024-04-13 09:00:00 | 2024-04-13 10:00:00 | 280 | 4
(3 rows)

(END)
baggage_check_id | check_result | created_at | updated_at | booking_id | passenger_id

1 | Not checked | 2024-04-10 12:00:00 | 2024-04-10 11:00:00 | 205 | 1
3 | Not checked | 2024-04-12 13:00:00 | 2024-04-12 14:00:00 | 250 | 3
4 | Rejected | 2024-04-13 09:00:00 | 2024-04-13 10:00:00 | 280 | 4
(3 rows)